Internal Portal Troubleshooting

Windows 11 Pro in Azure | Python 3.13.16 | Completed 23 October 2026

Independent lab using a fictional portal. Browser and server ran on the same VM; the server listened only on 127.0.0.1:8000.

My short interview answer

> "I built a small training portal and confirmed it worked by IP address and hostname. I then deliberately changed the local hosts entry to the wrong address. When the browser failed, I compared HTTP requests: the IP address returned 200 OK, but the hostname could not connect. That showed the server was still responding and helped narrow the investigation. I checked the resolved address and hosts file, and both showed the incorrect address. I corrected the entry, flushed the resolver cache, and verified recovery with a new HTTP request and the browser. I documented the findings and screenshot evidence. My main lesson was to test the service and name resolution separately before assuming a website or the internet is down."

Remember: Compare → Inspect → Correct → Verify

Compare	curl.exe returned 200 OK for 127.0.0.1:8000, while staffportal.test:8000 failed.
Inspect	The system resolver and hosts file both showed 127.0.0.2. Our server was listening at 127.0.0.1.
Correct	Changed only the added mapping back to 127.0.0.1 staffportal.test, saved it, then ran ipconfig /flushdns.
Verify	A new hostname request returned 200 OK, and Edge displayed the expected page. No server restart or Azure network change was needed.

Likely questions and simple answers

Was this a DNS server outage? No. It was an incorrect entry in the local hosts file. External DNS servers were not tested.

What does 200 OK mean? The server successfully responded to that HTTP request. I also checked the browser to verify the expected page content.

Why flush the cache? To clear stored resolver entries after changing the mapping. Flushing alone did not correct the wrong hosts entry.

What does 127.0.0.1 mean? It is a loopback address pointing to the computer itself. The server was bound specifically to that address.

Why bypass proxies in curl? To make these test requests direct and keep the IP-versus-hostname comparison consistent.

What would you do at work? Establish impact and scope, check ownership and the intended mapping, and make only authorised changes. Escalate shared, suspicious or centrally managed faults with evidence.

Explain the limits accurately

This did not demonstrate a wider network failure, multi-user outage, authenticated clinical application or HTTPS deployment. Python served a static training page. Its "Available" text was not live monitoring. The incident ticket was written retrospectively from lab evidence.

Tools: PowerShell, curl.exe, system name-resolution lookup, Select-String, Notepad and Edge.